

Ali TOURANI

SENIOR R&D SPECIALIST · COMPUTER VISION & GENAI EXPERT · FULL-STACK AI & SOFTWARE ENGINEER

Strassen, Luxembourg

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Scholar: [alitourani](https://scholar.google.com/citations?user=alitourani)

Summary

Senior Software Developer and Full-Stack AI Engineer with **10+ years** of experience architecting end-to-end **Applied AI, Robotics**, and **data-driven** systems from design to full-lifecycle deployment and maintenance. R&D expert in **Computer Vision, LLM/VLM, Embodied AI**, and **Multimodal RAG**, with **10+ open-source frameworks** (Python, C++, ROS) backed by cutting-edge research. Delivered **production-grade** web platforms across **5+ countries**. Experienced in **full-stack** engineering, **CI/CD** practices, **Docker Containerization**, and **Agile/Scrum** project management.

Work Experience

Interdisciplinary Centre for Security, Reliability, and Trust (SnT), Luxembourg

Lead R&D Specialist | Postdoctoral Researcher (full-time) | April 2026 – July 2026

- Led research on **Vision-Language Model (VLM)** grounding for robotic scene understanding, spatial reasoning, and knowledge retrieval.
- Designed and deployed an **embodied AI** core for spatial reasoning and **context-aware perception** in unstructured environments.
- Implemented a **Retrieval-Augmented Generation (RAG)** system with end-to-end **GenAI**-driven reasoning for real-world robotics.
- Mentored **three Ph.D. students** across system development, experimental design, and publication workflows.

Doctoral Researcher (full-time) | April 2022 – March 2026

- Developed various **computer vision** and **spatial AI** frameworks to improve autonomous robots' situational awareness.
- Designed, packaged, and maintained **5+ open-source** robotic software packages, adopted widely across industry and research teams.
- Authored **15+ publications** across top-tier scientific venues (journals and conferences), achieving an **h-index of 15**.

AVID Technology Development Co., Iran

Full-Stack Software Developer (full-time) | January 2019 – December 2021

- Built React/Node.js **full-stack** systems for **international** enterprise clients across **5+ countries**, as production-grade software delivery.
- Mentored junior developers and led **Agile/Scrum** sprints as secondary Scrum Master for **teams of 10+**.
- Implemented **CI/CD pipelines**, DevOps/MLOps workflows, and **Docker-based** deployments to support reliable production releases.

Guilan Science and Technology Park Business Incubator, Iran

Research Software Engineer (part-time) | July 2019 – July 2021

- Deployed two production-grade computer vision systems for **vehicle tracking** and **license plate recognition** in real-world traffic.
- Bridged research and industry by applying academic methods to scalable **Intelligent Transportation Systems (ITS)**.

ABF Varna Co., Iran

Software/Web Developer (full-time) | November 2016 - November 2017

- Designed intuitive **UI/UX interfaces** for web and mobile applications, focusing on responsive design and user-centric workflows.
- Collaborated within cross-functional teams, participating in code reviews for scalable software solutions.

IEHT Guilan Training Center, Iran

Interactive E-Learning Content Developer (freelance) | May 2013 - December 2016

- Developed high-impact interactive content using **ActionScript**, transforming technical concepts into digital learning experiences.

Education

Ph.D. in Computer Science (Doctorat en Informatique)

University of Luxembourg, Luxembourg | April 2022 – March 2026

- Thesis:** Enhancing Robots' Situational Awareness using Imperceptible Artificial Landmarks ([thesis link](#))
- Achievement:** Nominated for the **Excellent Thesis Award** of the University of Luxembourg (March 2026)
- Affiliation:** Automation and Robotics Research Group, Interdisciplinary Centre for Security, Reliability, and Trust (SnT) ([link](#))
- Supervisor:** Prof. Dr. Holger Voos ([link](#)) | **Advisor:** Dr. Jose Luis Sanchez-Lopez ([link](#))

M.Sc. Computer Engineering (Software Engineering Stream)

University of Guilan, Iran | September 2016 – January 2019

- Thesis:** Automatic Vehicle Speeding Violation Detection using Computer Vision Solutions ([link](#))
- Supervisor:** Prof. Asadollah Shahbahrami ([link](#))
- GPA:** 18.97 out of 20.00 (Excellent)

B.Sc. Computer Engineering (Software Engineering Stream)

University of Guilan, Iran | September 2009 – June 2013

- Thesis:** An E-Ticketing Software Platform for Appointment Scheduling in Medical Clinics

Selected Projects (open-source)

Sentra GenAI Toolkit (<https://github.com/snt-arg/sentra>)

- An **Embodied AI** framework employing **VLM** grounding for real-time **Multimodal RAG** and spatial **Visual Question-Answering**.
- **Stack:** Python, ROS, HuggingFace Transformers, Docker, PyTorch

Visual S-Graphs (vS-Graphs - https://github.com/snt-arg/visual_sgraphs)

- A robotic and **spatial AI** framework for **semantic scene understanding** and generating environments' interpretable **3D digital twins**.
- **Stack:** C++, Python, ROS, Colcon, CMake, OpenCV, Docker, TensorFlow

RAG-VisualRec Advanced Retrieval System (<https://github.com/recsys-lab/rag-visualrec>)

- A **Generative AI**-based resource for enhancing **reasoning** with visual, audio, and textual embeddings.
- **Stack:** Python, OpenAI/ST/Llama APIs, Notebook

Popcorn Video Recommendation Framework (<https://github.com/recsys-lab/popcorn>)

- Multimodal framework for **visual embeddings** extraction, fusion, augmentation, and **RAG**-based retrieval.
- **Stack:** Python, OpenCV, FFMpeg, Notebook

Vehicle & License Plate Detection/Tracking (<https://github.com/alitourani/yolo-lpd>)

- An end-to-end pipeline for **vehicle license plate detection and tracking** for smart barrier gates and vehicle **access control**.
- **Stack:** Python, C++, C#, OpenCV, YOLO, TensorFlow, Keras

Skills

Software Development

Desktop Stack

Web Stack

Databases

Extras

Expert: Python, C++ | **Proficient:** C# | **Familiar:** Java (Spring), Objective-C
Expert: React.js, Node.js, JavaScript, CSS, HTML | **Proficient:** Next.js, GraphQL, PHP | **Familiar:** Webpack
Expert: Microsoft SQL, MySQL | **Proficient:** PostgreSQL, MongoDB, SQLite | **Familiar:** Elasticsearch, Neo4j
Expert: Git, Linux, Jira | **Proficient:** Docker, REST API, .NET Framework | **Familiar:** CI/CD, VMware

Tools & Technologies

Computer Vision

Machine Learning

Robotics

Generative AI & LLM

Expert: OpenCV, YOLO | **Proficient:** CLIP, Scikit-Image | **Familiar:** EmguCV, AForge.NET
Expert: HuggingFace Transformers, Keras | **Proficient:** PyTorch, TensorFlow | **Familiar:** Scikit-learn
Expert: ROS2, RViz | **Proficient:** PCL, Voxblox | **Familiar:** GTSAM, Gazebo
Expert: VLMs, VLAs | **Proficient:** RAG, GPT | **Familiar:** Llama, SentenceTransformers

Soft Skills

Communication

Management

Teamwork

Languages

Presentation, Scientific Writing, Technical Documentation
Agile, Scrum, Mentoring (students, developers, interns), Project Coordination
Interdisciplinary collaboration, Team spirit, Cross-cultural collaboration
English (C1), French (B1), German (A2), Luxembourgish (A1), Turkish (A1), Persian (mother tongue)

Research Highlights

- Authored and co-authored **30+ peer-reviewed publications** in leading international conferences and journals across **Computer Vision**, **Robotics**, **Information Retrieval**, and **Embodied AI**.
- Peer reviewer for **top-tier scientific venues** and **national funding schemes**, evaluating state-of-the-art machine learning and AI methodologies, experimental design, and technical contributions.

The complete list of papers and preprints is available on my [Google Scholar](#) profile (**h-index: 15**).

Honors & Awards

- 2026** **Nomination**, Best Ph.D. Thesis Award Nominee, University of Luxembourg, Luxembourg
- 2022** **Scholarship**, Competitive Fully Funded Ph.D. Position (TRANSCEND Project), University of Luxembourg, Luxembourg
- 2019** **Award**, Best Student Researcher for Scientific and Practical Contributions (class of 2016), University of Guilan, Iran
- 2019** **Award**, Academic Excellence: Highest GPA in M.Sc. Computer Engineering (class of 2016), University of Guilan, Iran

Reference

Prof. Dr. Holger Voos (holger.voos@uni.lu)

Full Professor in Engineering Science, University of Luxembourg, Luxembourg (Ph.D. Supervisor)

Dr. Jose Luis Sanchez-Lopez (joseluis.sanchezlopez@uni.lu)

Senior Research Scientist, Automation & Robotics Group (ARG), SnT, Luxembourg (Ph.D. Advisor)

Dr. Arsham Mostaani (arsham.mostaani@losch.lu)

Senior Research & Innovation Strategist, Losch Digital Lab, Luxembourg (Research Collaboration)